The universal property of the circle: maps from S^1 to A are the based loops in A

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:S1ump' (source lines 330-335, Lemma 6.2.9); the proof environment begins on the very next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence, which states that the recursion and uniqueness lemmas for the circle imply this universal property; it is kept because the proof's second round trip is exactly the uniqueness principle.

Editorial note. The retained counter values (chapter 6, section 2, theorem offset 8) make the retained lemma print as Lemma 6.2.9, matching the original book numbering; the section-level theorem counter is shared by theorem, lemma and corollary through aliascnt.

Editorial note. No retained line uses a label, citation, figure, included file or non-ASCII character, so the package declares no external reference, no bibliography and no asset; the item compiles from the retained text alone.

Editorial scope. One original result of the HoTT Book, counted once: The universal property of the circle: maps from S^1 to A are the based loops in A.

Allowed standard prerequisites (editorial).

- Recursion principle of the circle. For a type A with a point a : A and a loop p : a = a there is f : S^1 -> A with f(base) := a and ap f (loop) = p; the non-dependent computation rule for loop is what turns the dependent datum into a path.. The inverse direction at source line 338 is defined by the circle recursion principle applied to the pair (b, l), and the first round trip at source line 340 is exactly its computation rule.
- Uniqueness principle for maps out of the circle. If f, g : S^1 -> A come with p : f(base) = g(base) and a dependent path over loop between ap f (loop) and ap g (loop), then f(x) = g(x) for all x : S^1.. The second round trip at source lines 341-342 is proved by this uniqueness principle, applied with p := refl(base); its hypothesis is discharged by the recursion computation rule.
- Quasi-inverses give equivalences. A function is an equivalence as soon as it has a two-sided homotopy inverse.. The two homotopies f o g ~ id and g o f ~ id at source lines 340-342 are exactly the data that the final sentence at source line 343 feeds into this characterization.

These two lemmas imply the expected universal property of the circle:

Lemma 6.2.9. *For any type A we have a natural equivalence*

$$(\mathbb{S}^1 \to A) \simeq \sum_{x:A} (x = x).$$

Proof. We have a canonical function $f : (\mathbb{S}^1 \to A) \to \sum_{(x:A)}(x = x)$ defined by $f(g) :\equiv (g(\mathsf{base}), g(\mathsf{loop}))$. In the other direction, we have $g : \sum_{(x:A)}(x = x) \to (\mathbb{S}^1 \to A)$ defined by taking a pair (b, ℓ) to the function $\mathbb{S}^1 \to A$ given by the recursion principle of the circle.

Now, by the computation rule of the recursion principle, $f \circ g \sim \mathsf{id}$. Whereas $g \circ f \sim \mathsf{id}$ by the uniqueness principle, since $(g \circ f)(\mathsf{loop}) =^{\lambda x.\, x =_A x}_{\mathsf{refl}_{\mathsf{base}}} \mathsf{loop}$, again, by the computation rule of the recursion principle of the circle. Thus, f has a quasi-inverse, and is therefore an equivalence. □